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Course: Java FSE Digital Nurture

Week 1

Exercise 1: Logging Error Messages and Warning Levels Using SLF4J

Aim

To understand and implement logging of error and warning messages using the SLF4J logging framework.

Objective

The objective of this exercise is to configure SLF4J with Logback and demonstrate logging at different severity levels such as ERROR and WARN.

Software Requirements

- Java Development Kit (JDK)
 - Apache Maven
 - Visual Studio Code
 - SLF4J
 - Logback
-

Theory

SLF4J (Simple Logging Facade for Java) is a logging abstraction framework that allows developers to use different logging implementations without changing application code.

Logback is a popular implementation of the SLF4J API.

Logging helps developers:

- Monitor application behavior.
- Track warnings and errors.
- Debug issues efficiently.
- Maintain application logs.

Logging Levels

ERROR

Used for serious problems that may prevent the application from functioning correctly.

Example:

```
logger.error("This is an error message");
```

WARN

Used to indicate potentially harmful situations that do not immediately stop execution.

Example:

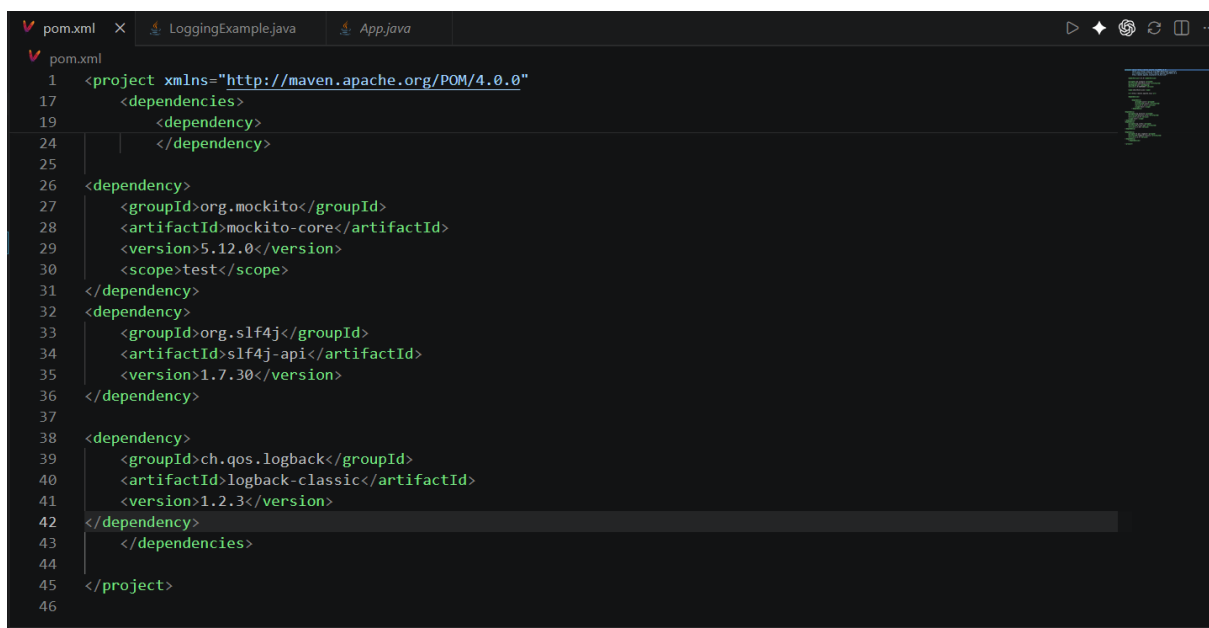
```
logger.warn("This is a warning message");
```

Procedure

Step 1: Add Dependencies

The SLF4J API and Logback Classic dependencies were added to the Maven project's pom.xml file.

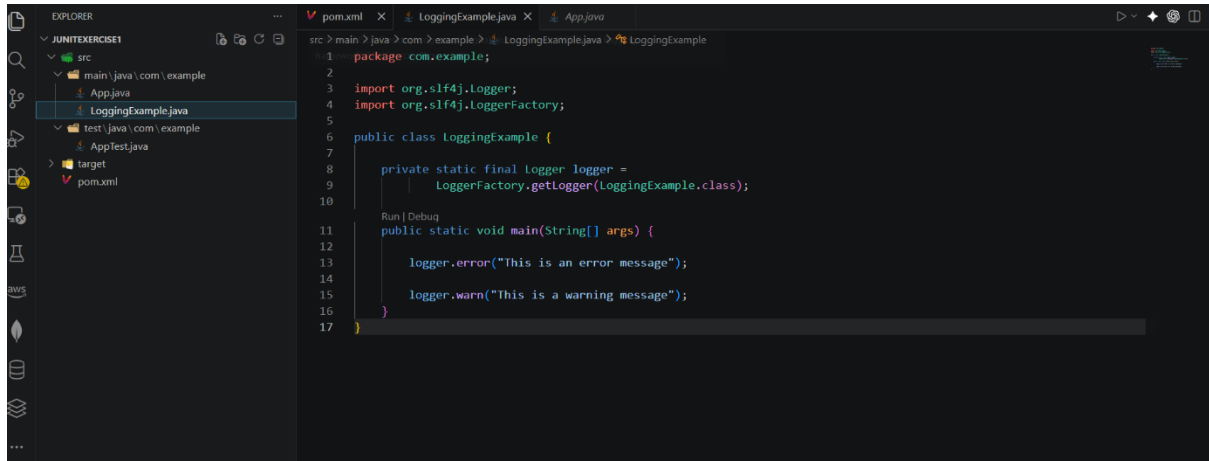
Screenshot 1:



Step 2: Create Logging Class

A Java class named `LoggingExample.java` was created to demonstrate logging functionality.

Screenshot 2:



Step 3: Implement Logging Statements

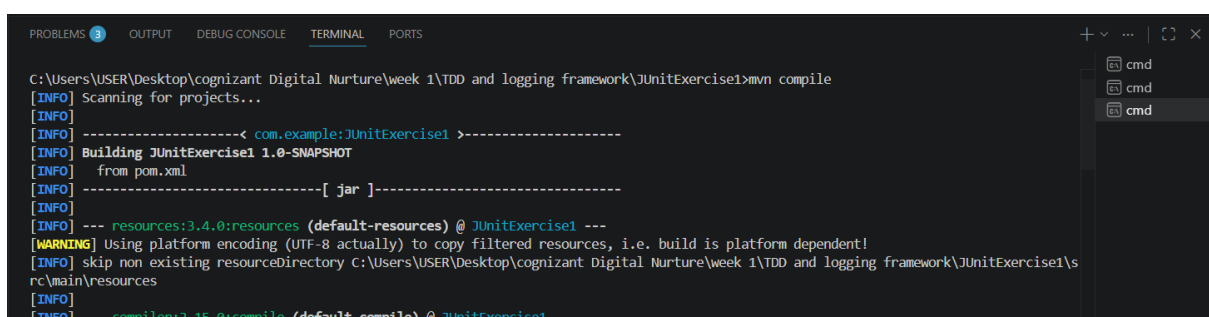
The logger object was created using `LoggerFactory` and used to print `ERROR` and `WARN` messages.



Step 4: Execute Program

The application was executed using Maven and the log messages were displayed in the console.

Screenshot 4:



Output

The application produced the following log messages:

ERROR com.example.LoggingExample - This is an error message

WARN com.example.LoggingExample - This is a warning message

Result

The SLF4J logging framework was successfully configured and used to generate error and warning messages.

Conclusion

This exercise demonstrated the integration of SLF4J with Logback for logging purposes. Logging levels such as ERROR and WARN were successfully implemented and displayed, helping in monitoring and debugging application behavior.